KEVIN SCHWARZWALD

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EDUCATION

Doctor of Philosophy, Earth and Environmental Sciences October 2024
Master of Philosophy, Earth and Environmental Sciences October 2023
Master of Arts, Earth and Environmental Sciences May 2021
Columbia University in the City of New York New York, NY

Research Areas: climate variability, uncertainty in projections of climate and climate impacts, precipitation in East Africa, tropical convection, climate model evaluation, the use of climate projections in policy and adaptation efforts
Advisors: Lisa Goddard (†2022), Richard Seager; *Committee Members*: Mingfang Ting, Kate Marvel, Radley Horton

- PhD thesis: *The Present and Future of the Horn of Africa Rains*. Link.

Master of Law in China Studies (concentration in Politics and International Relations) July 2017
Peking University, Yenching Academy (北京大学燕京学堂) Beijing, China

Honors: Awards for Outstanding Academic Achievement and Outstanding Contributions to the Academy

- Master's thesis: *The Relationship between Fiscal Systems and Inefficient Urban Expansion in the PRC: An Empirical Look at Prefecture-Level Urbanization*

Bachelor of the Arts (A.B.) in Physics; Public Policy (concentration in Economics) June 2015
The University of Chicago Chicago, IL

Honors: Graduated with General Honors, Departmental Honors in Public Policy

- Senior Thesis (Public Policy): *"The Wrong Folks Kept Winning": Protectionism and America's Trains*

ACADEMIC RESEARCH EXPERIENCE

University of California, Davis Starting September 2026
Postdoctoral Scholar Davis, CA and remotely from New York, NY
Principal Investigators: Fran Moore, Adam Sobel, and Michael Tippett

Columbia Climate School September 2024 – August 2026
Climate School Postdoctoral Scientist New York, NY
Mentors: Radley Horton and Gernot Wagner
(On paternity leave Jan-Feb and Apr-Jun 2025)

Lamont-Doherty Earth Observatory (LDEO), September 2019 – August 2024
International Research Institute for Climate and Society (IRI) New York, NY
Graduate Research Assistant
Advisors: Lisa Goddard (†2022), Richard Seager

Energy Policy Institute at the University of Chicago (EPIC) September 2017 – July 2019
Pre-Doctoral Fellow (joint with RDCEP) Chicago, IL and remotely from New York, NY as NASA GISS Visiting Researcher
Principal Investigator: Amir Jina

Center for Robust Decision-Making on Climate and Energy Policy (RDCEP) April 2015 – July 2019
Pre-Doctoral Fellow (joint with EPIC 2017-2019) Chicago, IL and remotely from New York, NY, and Beijing, China
Principal Investigator: Elisabeth J. Moyer

PUBLICATIONS (under review and upcoming)

Schwarzwald, Kevin, Nathan Lenssen, Tom Bearpark, Alia Bonanno, Radley Horton, and Gernot Wagner. "Observational uncertainty is the primary source of climate uncertainty in impacts projections" *Under review at Science Advances.*

Schwarzwald, Kevin, Nathan Lenssen, Radley Horton, and Gernot Wagner. "A bias-corrected & downscaled massive ensemble to diagnose uncertainty in climate impact projections." *Under revision at Scientific Data.* Link.

Mauerman, Max, **Kevin Schwarzwald**, Daniel Osgood, Nathan Lenssen, and Abdel-Lathif Younous. "Sampling Uncertainty is Critical for Setting Forecast-Based Anticipatory Action Triggers" *Under revision at Bulletin of the American Meteorological Society.*

Lenoble, Colin, Mathias Valla, Jonas Schmitt, Céline Guivarch, Améline Vallet, Pauline Rivoire, **Kevin Schwarzwald**, and Kai Kornhuber. "Global risks of renewable energy droughts under climate change" *Under revision at Nature Communications.*

Schwarzwald, Kevin and Nathan Lenssen. *Solicited textbook chapter, in progress*. "The Sources of Uncertainty in Climate Projections." in *Encyclopedia of Climate System Science*, edited by Sumant Nigam and Isla R. Simpson. Elsevier Major Reference Works. Elsevier.

Kisembe, Jesse, Yixin Wen, Chris Funk, **Kevin Schwarzwald**, Ronald I. Odongo, Weikang Qian, and Samuel Ekwacu. "Asymmetric Indo-Pacific Drivers Produce Later Shorter Seasons with Earlier Longest Dry Spells or Earlier Longer Seasons with Later Longest Dry Spells." *Submitted to Journal of Hydrometeorology*.

PUBLICATIONS (peer-reviewed)

Schwarzwald, Kevin and Kerrie Geil. 2024. "`xagg`: A Python Package to Aggregate Gridded Data onto Polygons." *Journal of Open Source Software* 9 (104): 7239. Link.

Schwarzwald, Kevin and Richard Seager. 2024. "Revisiting the 'East African Paradox': CMIP6 models also struggle to reproduce strong observed MAM drying trends." *Journal of Climate* 37 (24): 6641-58. Link.

Rising, James A., Azhar Hussain, **Kevin Schwarzwald**, and Ana Trisovic. 2024. "A Practical Guide to Climate Econometrics: Navigating Key Decision Points in Weather and Climate Data Analysis." *Journal of Open Source Education* 7 (75): 90 Link.

Schwarzwald, Kevin, Richard Seager, Mingfang Ting, and Alessandra Giannini. 2023. "Large-Scale Stability and the Greater Horn of Africa Long and Short Rains." *Journal of Climate* 36 (20): 7297–7317. Link.

Anderson, Weston, Benjamin I. Cook, Kim Slinski, **Kevin Schwarzwald**, Amy McNally, and Chris Funk. 2023. "Multiyear La Niña Events and Multiseason Drought in the Horn of Africa." *Journal of Hydrometeorology* 24 (1): 119–31. Link.

Schwarzwald, Kevin, and Nathan Lenssen. 2022. "The Importance of Internal Climate Variability in Climate Impact Projections." *Proceedings of the National Academy of Sciences* 119 (42). Link.

Schwarzwald, Kevin, Lisa Goddard, Richard Seager, Mingfang Ting, and Kate Marvel. 2022. "Understanding CMIP6 Biases in the Representation of the Greater Horn of Africa Long and Short Rains." *Climate Dynamics* 61 (3): 1229–55. Link.

Schwarzwald, Kevin, Andrew Poppick, Maria Rugenstein, Jonah Bloch-Johnson, Jiali Wang, David McInerney, and Elisabeth J. Moyer. 2021. "Changes in Future Precipitation Mean and Variability across Scales." *Journal of Climate* 34 (7): 2741–58. Link.

CONFERENCE PRESENTATIONS

* indicates presenting author

Schwarzwald, Kevin*, Nathan Lenssen, Radley Horton, Alia Bonnano, and Gernot Wagner. 2026. "The choice of historical data product dominates climate uncertainty in projections of climate impacts in a 2-degree world" *Highlighted presentation at the EGU General Assembly*, Vienna, Austria.

Schwarzwald, Kevin*, Noel Siegert, Nathan Lenssen, Radley Horton, and Patrick Alexander. 2025. "Discrepancies in near-term projections of climate impacts due to persistent issues in simulated SST trends in GCMs" *Presentation at the AGU Fall Meeting*, New Orleans, LA.

Schwarzwald, Kevin*, Nathan Lenssen, Radley Horton, Alia Bonnano, and Gernot Wagner. 2025. "Internal variability, observational uncertainty substantially contribute to uncertainty in global projections of heat-related mortality in a 2-degree world: Insights from bias-corrected Large Ensembles." *Presentation at the AGU Fall Meeting*, New Orleans, LA.

Schwarzwald, Kevin*. 2025. "Climate uncertainty in the range of future climate risks." *Invited keynote presentation at the MLISTRAL 2 meeting on Climate Change and Insurability,* Marseille, France.

Lenssen, Nathan* and **Kevin Schwarzwald**. 2025. "Climate Projection Uncertainty of Future Heat Extremes." *Invited oral presentation at the Contemporary Advances in Statistics of Extremes Workshop,* Columbia, MO.

Schwarzwald, Kevin* and Richard Seager. 2024. "Revisiting the 'East African Paradox': CMIP6 models also struggle to reproduce strong observed MAM drying trends." *PICO Presentation at the EGU Annual Meeting*, Vienna, Austria.

Schwarzwald, Kevin* and Nathan Lenssen. 2023. "On the propagation of climate data uncertainty in projections of the impacts of climate change on society." *Invited oral presentation at the AGU Fall Meeting*, San Francisco, CA.

Schwarzwald, Kevin* and Richard Seager. 2023. "Revisiting the 'East African Paradox': Long Rain trends and variability in models and observations in the Horn of Africa." *Poster at the AGU Fall Meeting*, San Francisco, CA.

Schwarzwald, Kevin* and Nathan Lenssen. 2023. "Understanding the sources of climate uncertainty in projections of climate impacts." *Poster at the WCRP Open Science Conference,* Kigali, Rwanda.

Schwarzwald, Kevin*, Richard Seager, Mingfang Ting, and Kate Marvel. 2023. "Biases in simulations of the Horn of Africa long and short rains in global climate models and their implication for future rainfall changes." *Poster at the WCRP Open Science Conference,* Kigali, Rwanda.

Schwarzwald, Kevin* and Nathan Lenssen. 2023. "Understanding the sources of climate uncertainty in projections of climate impacts." *Presentation at the EGU Annual Meeting,* Vienna, Austria.

Schwarzwald, Kevin*, Richard Seager, and Mingfang Ting. 2023. "The Moist Static Energy Seasonal Cycle and the Horn of Africa Long and Short Rains." *Presentation at the 103rd AMS Annual Meeting,* Denver, CO.
- Awarded Outstanding Oral Presentation Award

Schwarzwald, Kevin*, and Nathan Lenssen. 2022. "A Complete Assessment of Climate Uncertainty in Projections of Climate Impacts." *Poster at the AGU Fall Meeting*, Chicago, IL.

Schwarzwald, Kevin*, and Nathan Lenssen. 2022. "A Complete Assessment of Climate Uncertainty in Projections of Climate Impacts." *Poster at the Institute for Mathematical and Statistical Innovation (IMSI) Workshop on "Economic Impacts of Climate Change"*, Chicago, IL.

Schwarzwald, Kevin*. 2022. "The sources of climate uncertainty in climate impacts studies." *Presentation at the Interdisciplinary PhD Workshop in Sustainable Development (IPWSD)*, New York, NY.

Lenssen, Nathan, and **Kevin Schwarzwald***. 2021. "Uncertainty in Climate Impacts Projections Due to Model Internal Variability." *Presentation at the AGU Fall Meeting*, New Orleans, LA

Schwarzwald, Kevin*, Richard Seager, Kate Marvel, Mingfang Ting, and Lisa Goddard. 2021. "Diagnosing CMIP6 Model Biases in East African Rainfall." *Poster at the AGU Fall Meeting*, New Orleans, LA

Schwarzwald, Kevin*, Matz A Haugen, Elisabeth J Moyer, and Amir Jina. 2018. "Economic Impacts of Variability under Climate Change on Mortality Projections" *Presentation at the US CLIVAR Large Ensembles Workshop,* Boulder, CO.

Schwarzwald, Kevin*, Amir Jina, Matz A Haugen, Michael Stein, and Elisabeth J Moyer. 2018. "Integrating Changes in Temperature Variability into Climate Damage Projections." *Poster at the AGU Fall Meeting*, Washington, DC.

Schwarzwald, Kevin*, Andrew N Poppick, and Elisabeth J Moyer. 2018. "Frequency-Dependent Changes in Temperature Variability across Global Climate Models." *Poster at the AGU Fall Meeting*, Washington, DC.

Moyer, Elisabeth J*, **Kevin Schwarzwald**, Won Chang, David McInerney, Whitney K Huang, and Victor Zhorin. 2016. "Changes in Interannual and Interdecadal Precipitation Variability from Millennial-Scale Climate Model Runs." *Poster at the AGU Fall Meeting*, San Francisco, CA.

SELECTED INVITED SEMINARS

"Climate uncertainties in projections of climate impacts." *Mailman School of Public Health*. March 23, 2026.

"Climate uncertainties in projections of climate impacts." *International Institute for Applied Systems Analysis (IIASA)*. March 16, 2026.

"Two puzzles in understanding the Horn of Africa long and short rains." *Oxford School of Geography and the Environment Seminar*. September 20, 2024.

"Updating the East African Paradox: Inconsistent Rainfall Trends between CMIP6 Models and Observations." *Max-Planck-Institut für Meteorologie Joint Seminar*. July 24, 2024.

"Revisiting the 'East African Paradox': Why do GCMs project increased MAM rainfall?" *WCRP EPESC Science Meeting*. May 16, 2024.

"On the propagation of climate data uncertainty in projections of the impacts of climate change on society." *JPL Science Understanding through Data Science (SUDS) Seminar Series*. January 23, 2024.

"What we know (and don't know) about the future of East African rainfall." *IRI PyCPT Training for 30 visiting scientists from Meteorological Offices across Africa*. September 15, 2023.

"On the propagation of climate data uncertainty in projections of the impacts of climate change on society." *ECS Symposium*, May 22, 2023. Link.

"The Horn of Africa Rains and How We Model Them." *UCSB Climate Hazards Center Climate Meeting*, February 10, 2023.

"The sources of climate uncertainty in climate impacts studies." *NASA GISS Lunch Seminar*, February 23, 2022. Link.

"Climate Uncertainty in Projections of the Impacts of Climate Change." *Columbia Student Climate Symposium*, February 18, 2022.

"Process-based climate model evaluations in the Horn of Africa." *UChicago Environmental Data Lunch*, March 22, 2021.

AWARDS, GRANTS, COMPETITIVE PROGRAMS

- Climate School Postdoctoral Program; successful institutional fellowship application for "*The Full Magnitude of Climate Uncertainty in Projections of Climate Impacts: A Case Study in Tropical, Developing Countries*", **$145,430** (salary) — 2024-2026
- CATER School on Transdisciplinary Climate Risk Adaptation and Action (Naivasha, Kenya; fully funded, including travel) — 2023
- ASciNA Young Scientist Award, **$7500** — 2023
- Outstanding Oral Presentation Award, 103rd AMS Annual Meeting — 2023
- Center for Science and Society Seed Grants, **$3975** total (to NENSIC, co-PI) — 2021, 2022, 2023
- ICTP Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics (Trieste, Italy; fully funded) — 2022
- Yenching Academy Dean's Research Grant, **¥5000** — 2016

TEACHING AND MENTORING EXPERIENCE

Project Supervisor — September 2024 – present
3rd Como Summer School on Compound Events

Managing and Mentoring
Milind Kudapa (graduate research assistant) — June 2026 – present
Alia Bonanno (undergraduate research assistant) — June 2025 – June 2026
Colin Lenoble (graduate student at CIRED, France) — September 2024 – present

Graduate Teaching Assistant
EESCW2100: *Earth's Environmental Systems: Climate Systems* — January 2023 – March 2023
- Ran office hours, led labs and field trips, graded assignments for an undergraduate intro climate science course in Columbia's Dept. of Earth & Environmental Sciences, taught by Drs. Mingfang Ting and Gisela Winckler
- Evaluations of teaching effectiveness: Excellent (10/13 respondents), Very Good (3/13)

CLMTGR5001: *Dynamics of Climate Variability and Change* — September 2020 – December 2020
- Ran virtual office hours, led discussions, graded assignments for an intro climate science course in Columbia Climate School's interdisciplinary Climate and Society Master's, taught by Drs. Alessandra Giannini and Weston Anderson
- Taught a lecture on the drivers of climate uncertainty and the interpretation of climate model output
- Evaluations of teaching effectiveness: Excellent (11/14 respondents), Very Good (3/14)

Columbia Summer School in the Green Mountains Pre-College Program
Instructor — June 2022
- Developed course material and taught lectures on future climate projections, climate variability, and the epistemology of climate science for a high school summer program

Additional Invited Class Lectures
- "Climate and climate impact uncertainty," for *Applied Climate* in Columbia Climate School's Climate & Society Master's Program (*Spring 2026*)
- "Understanding climate models and future climate change," for *Climate Risk Management* at Virginia Tech (*October 2023*)
- "Climate models and projections," for *Dynamics of Climate Variability and Change* in Columbia Climate School's Climate & Society Master's Program, in a flipped classroom setting (*November 2023 and November 2024*)

ACADEMIC SERVICE, EDUCATION, AND OUTREACH

PLOS Climate — July 2026 – present
Guest Editorial Board Member

WCRP EPESC Lighthouse Activity, East Africa Case Study — November 2023 – present
Interim Steering Committee Member

History and Climate Change Workshop and Seminar Series — January 2022 – May 2023
Student Convener

Lamont-Doherty Earth Observatory Executive Committee — September 2021 – August 2022
Graduate Student Representative

IRI's Climate Society Community Outreach (CSCO) group July 2020 – 2022

- Participating in the IRI's education and outreach group, formed in response to the 2020 U.S. protests for racial justice
- Mentored New York, NY high school students in data science projects through the "Make With Data" project
- Co-taught class sessions on climate variability at Hamilton Grange Middle School in Harlem, New York, NY

New Era Network for Societally Integrated Climatology (NENSIC) May 2020 – August 2025

Co-founder, Organizer *Website*: nensic.org

- Launched an interdisciplinary network designed to connect young researchers, students, and professionals working on climate issues in any field, with a focus on improving interdisciplinary communication, building informal connections, peer mentoring, and providing a welcoming, inclusive venue to present work and find collaborators
- Organized and co-organized 60+ events in person and remotely with participants from four continents, with research talks on topics such as international climate negotiations, writing public comments as a scientist, or the coordination of disaster planning in Ecuador, in addition to networking events, tutorials, and interdisciplinary job panels

Reviews

- Refereed for *Nature Climate Change; Nature Communications; npj Climate and Atmospheric Science; Proceedings of the National Academy of Sciences (PNAS); Earth's Future; Environmental Research: Climate; Journal of Climate; Geophysical Research Letters; International Journal of Climatology; Weather & Climate Dynamics; Weather, Climate & Society; Geoscientific Model Development; Journal of Open Source Software; Environmental Modelling & Software; Climate Research; Atmósfera; Climatic Change; AAR2 – Austrian Assessment Report on Climate Change*

Media

- Featured in *Salzburger Nachrichten*, *Österreichischer Rundfunk (ORF), Kurier, Austria Presse Agentur (APA)*

Textbook Chapters

- **Schwarzwald, Kevin**. *In progress*. "The Sources of Uncertainty in Climate Projections." in *Encyclopedia of Climate System Science*, edited by Sumant Nigam and Isla R. Simpson. Elsevier Major Reference Works. Elsevier.
- **Schwarzwald, Kevin**. 2020. "Weather and Climate Data." in *A Practical Guide to Climate Econometrics: Navigating Key Decision Points in Weather and Climate Data Analysis*, edited by James Rising, Azhar Hussain, Kevin Schwarzwald, and Ana Trisovic. **climateestimate.net**.

OTHER SELECTED EXPERIENCE

Eparque Urban Strategies March 2019 – July 2019

Independent Consultant (part-time) New York, NY

Yenching Global Symposium (全球青年中国论坛) October 2015 – April 2016

Associate Director, Logistics Beijing, China

INVITED AND CONTRIBUTED PRESENTATIONS

2026: International Institute for Applied Systems Analysis (IIASA), Mailman School of Public Health, EGU General Assembly (**highlighted talk**), NYU Institute for Policy Integrity
2025: MLISTRAL2 Meeting (**invited keynote**), American Geophysical Union (AGU) Fall Meeting
2024: JPL Science Understanding through Data Science (SUDS) Seminar; European Geosciences Union (EGU)'s General Assembly, Max Planck Institute for Meteorology Hamburg, Oxford School of Geography and the Environment
2023: American Meteorological Association (AMS)'s Annual Meeting; EGU General Assembly; University of California Santa Barbara (UCSB) Climate Hazards Center; ECS Symposium; WCRP Open Science Conference; AGU Fall Meeting (**invited abstract**)
2022: AGU Fall Meeting; Institute for Mathematical and Statistical Innovation (IMSI); NASA Goddard Institute for Space Studies (GISS); ICTP Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics; Interdisciplinary PhD Student Workshop in Sustainable Development (IPWSD); Columbia Student Climate Symposium
2021: AGU Fall Meeting; University of Chicago Environmental Data Science Lunch
2019: US CLIVAR Large Ensembles Workshop
2018: AGU Fall Meeting

DATA TOOLS AND TUTORIALS

Bias-Corrected and Downscaled Massive Ensemble (BCD-ME) github.com/ks905383/bcd_me

Link to preprint *(links to data access in readme at above link)*

- A dataset of over 1400 projections of daily mean and maximum temperature from up to 86 ensemble members of 12 Global Climate Models (GCMs), bias-corrected and downscaled to four reanalysis products, organized by global

warming level, designed to test the sensitivity of climate projections and climate impact projections to all sources of climate uncertainty. The BCD-ME is freely available in Analysis-Ready, Cloud-Optimized (ARCO) Zarr format

xagg github.com/ks905383/xagg

- Lead developer and maintainer of **xagg**, a python package with over 30,000 cumulative downloads for aggregating raster data (such as gridded weather data) onto polygons (such as administrative shapefiles), using area-averaging based on the overlap between pixels, polygons

Weather Panel Tutorial climateestimate.net

- In an inter-disciplinary team, co-developed a tutorial designed to introduce climate economists to working with observational climate data using python (xarray), R, or MATLAB

quantproj github.com/ks905383/quantproj

- Developer of **quantproj**, an R package for projecting weather data by using downscaled fine-scaled distributional changes in Large Ensembles and the delta method; based on the methodology of Haugen et al., 2018

Atlas of Variability github.com/RDCEP/atlas-of-variability-code-package

- Created a MATLAB package to analyze variability statistics in global climate models
- The package was used to create the 'Atlas of Variability' – a publicly-accessible dataset of basic characteristics of climate variability for 20 commonly-analyzed variables in CMIP5-generation models

SKILLS

Analysis: advanced proficiency in python (xarray, dask), MATLAB, R, and Excel, basic proficiency in GIS (QGIS) and Stata, experience with climate (NetCDF, zarr, nco, CMIP, etc.) and satellite data analysis (DMSP/OLS, VIIRS, etc.)
High-performance computing: proficiency with work on HPC systems, including working with schedulers, writing optimized and parallelized code, etc., proficiency with NCAR's HPC and file storage system
Modeling: experience in running SPEEDY, an intermediate-complexity global climate model (GCM)
Communication/Design: advanced proficiency in LaTeX, PowerPoint, Keynote, inkscape; basic proficiency in Adobe Illustrator
Language: native in German, fluent in French, basic communication skills in Mandarin Chinese

PROFESSIONAL NETWORKS

- American Geophysical Union (AGU)
- European Geosciences Union (EGU)
- Austrian Scientists & Scholars in North America (ASciNA)